

Is it a bird? Is it a plane? Yep it is.

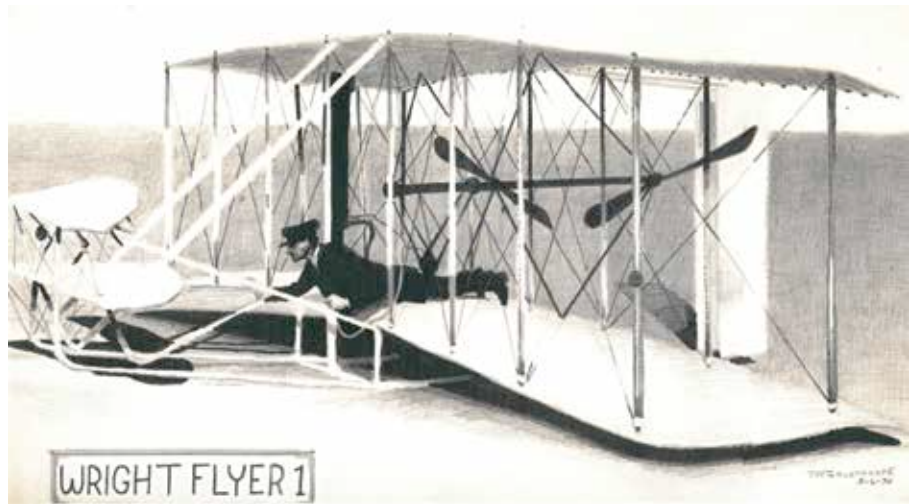
A Look At the Aircraft Designs Of The Future

Abhishek Choudhary

readersletters@thinkdigit.com

Futuristic aircraft, the way they are presented in the cartoons and movies is far from realistic. But that is not to say that the future of manned flight would not be radically different from what we know of them today. On the right is what we started out with, when the world was taking its first steps in our journey to the heavens. Not to put the design or the creators down by any measure, but Wright brothers did not have the best of designs for sustained flying. Come to think of it, long flight wasn't their purpose, either - they just needed to show that it could be done, in order to spur the rest of the world into action. And spur into action it did.

Now look at the image below. This is what the majority of mankind thinks of, when words like "plane", "flight" or "air travel" are mentioned. And no wonder, either (sorry helicopters, but we'll come back at you later), because as it turns out, aircraft that are roughly based on this design in the world dwarf the ones which aren't by something like 5:1. These are incredibly useful machines, but these designs aren't really the most efficient. Philosophically speaking, they accurately reflect everything that is wrong with the world, and everything that is leading us to, among other things, supremely high fuel prices. The basic idea here seems to be that if you throw enough engine power at something, it will fly (and fly it does). So we just attach super-high-powered engines at



We've come a long way from this haven't we?

something that will create an obstruction in the path of air, and air will essentially toss it up. This is the same principle by which vehicles, homes and – in a surprisingly high number of cases – cows fly in the event of hurricanes. Doesn't paint a pretty picture, does it now?

What the future holds

It doesn't really take much of a genius in aircraft design to figure out that it is really only the wings of the general aircraft that do the flying. The rest of the body just tries to keep as much out of the air's way as it can. There it is - Aerodynamics 101. So obviously, when tasked with producing more efficient aircraft design, the first shot will be at eradicating the notion of the wings and the body in a plane - just one huge, flying wing. And to think that this is a new design concept would be a huge disrespect to the man who pioneered the idea as early as 1910; one Mr. Hugo Junkers. A German engineer by trade, he quickly realised that a clean flying wing is the most aerodynamically efficient design configuration, and therefore the best way forward (being an intelligent man, he patented the design too). The issues at the time, of stability (or lack, thereof) and impossible navigation requirements, have now

been tackled effectively with the advent of drive-by-wire techniques in flight, which is just fancy talk for computer aided flight. That is the reason the flying wings have made a respectable comeback in the race to become the core design for the future, in addition to being already operational in some capacity for many years now. The future definitely looks good for this design in the subsonic arena, at least.

Of course, as a parallel to the flying wing design thought-chain, we could say that if, for some reason, we want to avoid making everything one single wing based-design, we could just go on adding the number of wings on our current models. This seems to be the motivation behind the boxed wing concept, where a huge, wing-based structure encapsulates the entire basic aircraft structure, and its edges all act like separate wings. In addition to reducing the drag of the overall configuration, it handles the sta-



Usually "Airplane" means this



The flying wing in the commercial space